

POLICY RESEARCH DOCUMENT

Regulating Algorithmic

Amplitude:

A Cognitive Stability

Framework

A Theoretical and Regulatory Model for Managing

AI-Driven Cognitive Perturbation at Scale

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Date: July 2026

Table of Contents

Section	Title	Page
1	Executive Summary	3
2	Glossary of Technical Terms	5
3	Scientific Foundations	7
4	Mathematical Models	11
5	Visual Diagrams	18
6	Risk Analysis	27
7	Regulatory Rationale	30
8	Safe Operating Range Specification	33
9	Governance Structure	36
10	Implementation Guidelines	39
11	Conclusion	43
A	Appendix — Notation Summary Table	45

1. Executive Summary

1.1 The Problem

Modern algorithmic content delivery systems now operate across billions of users simultaneously, generating and distributing content at a scale and precision unprecedented in the history of human communication. These systems do not merely distribute information — they actively curate, sequence, and intensify cognitively stimulating material in real time, producing what this framework terms **Algorithmic Amplitude (AA)**: the combined measure of the magnitude, velocity, frequency, and targeting precision of cognitively engaging or destabilizing content delivered to an individual or population over a defined time window. Unlike classical broadcast media, which exerted relatively uniform perturbation across populations, contemporary algorithmic systems possess the technical capacity to finely tune content intensity to individual cognitive profiles, creating conditions in which human cognitive equilibrium is not merely challenged but systematically and continuously perturbed at extraordinary scale.

The implications of this shift are profound and underappreciated. Human cognition is not a passive receiver of information but an active dynamical system that maintains stability through a set of self-regulating processes — epistemic damping, social feedback, and metacognitive self-monitoring. These processes evolved in environments characterized by relatively low information velocity and moderate affective intensity. Algorithmic amplitude, by contrast, can deliver content at intensities, velocities, and precision levels that far exceed the natural damping capacity of the human cognitive system. The result is not merely cognitive overload in the colloquial sense but a formal displacement of cognitive state vectors away from their stable attractor basins toward configurations characterized by polarization, conspiratorial ideation, hypervigilance, or learned helplessness. This document provides the conceptual vocabulary and mathematical scaffolding to analyze these effects with the rigour they demand.

1.2 Why It Matters

Without regulatory intervention, high-amplitude algorithmic systems are capable of progressively shifting population-level cognition away from stable attractor states toward what this framework designates **Negative Attractor Basins** — stable but pathological cognitive configurations that, once captured, resist correction through ordinary epistemic

means. The mechanisms of this shift are not accidental; they are the foreseeable outputs of engagement-optimized systems that systematically favor high-amplitude content because such content generates the affective arousal associated with extended platform engagement. The costs of this process are borne not by the platform operators who profit from it but by the broader social fabric that depends upon a stable epistemic commons — a shared cognitive baseline that makes collective decision-making, democratic deliberation, institutional trust, and cooperative norm-maintenance possible. When population-level cognitive distributions drift toward destabilized states, these shared goods erode in predictable and measurable ways: political polarization increases, institutional trust declines, conspiracy belief proliferates, and the capacity for collective rational action diminishes. These are not hypothetical harms — they are the observable social sequelae of unconstrained amplitude optimization at scale.

1.3 The Framework

This document proposes the **Cognitive Stability Framework (CSF)**, a comprehensive theoretical and regulatory model designed to establish safe operating ranges for algorithmic amplitude and provide governance mechanisms for their enforcement. The CSF comprises four interlocking components: (i) a formal mathematical specification of algorithmic amplitude and its components, enabling cross-platform measurement and comparison; (ii) a dynamical systems model of human cognitive stability, including attractor basin definitions, basin erosion dynamics, and population-level stability indices; (iii) a tiered regulatory threshold structure defining Green, Amber, and Red operating zones calibrated to empirically grounded epistemic damping estimates; and (iv) a governance architecture specifying an independent oversight body, a scientific advisory panel, and mandatory transparency and reporting requirements for covered platforms. Together, these components provide the conceptual, mathematical, and institutional foundations required to bring algorithmic amplitude within the domain of responsible regulatory oversight.

1.4 Call to Action

The urgency of this framework cannot be overstated. Basin erosion — the progressive weakening of a cognitive system's capacity to resist displacement from its primary attractor — is a cumulative and potentially irreversible process. Each interval of high-amplitude exposure without corrective intervention incrementally reduces the residual damping capacity of the affected cognitive system, lowering the effective amplitude threshold required to induce basin crossing in future exposures. At the population level, this means that the longer high-amplitude algorithmic systems operate without constraint, the more difficult and costly restoration of cognitive stability becomes. Regulators, platform operators, and scientific advisory bodies must therefore act collectively and without delay

to establish amplitude standards, measurement protocols, and enforcement mechanisms. The Cognitive Stability Framework provides the architecture for that action. Its adoption is not merely a policy preference — it is a prerequisite for the preservation of the epistemic foundations upon which stable, democratic, and cooperative societies depend.

2. Glossary of Technical Terms

The following definitions establish the precise technical vocabulary used throughout this framework. All terms are original to the Cognitive Stability Framework unless otherwise indicated. Readers are advised to familiarise themselves with this glossary before proceeding to the mathematical and analytical sections.

Term	Definition
Algorithmic Amplitude (AA)	The combined measure of the magnitude, velocity, frequency, and targeting precision of cognitively stimulating content delivered by an algorithmic system to a user or population over a defined time window. AA is the primary regulatory quantity of the CSF.
Cognitive Attractor State	A stable configuration of beliefs, emotional responses, and cognitive patterns toward which a mind naturally returns after perturbation, analogous to a physical system's energy minimum in a potential energy landscape. Attractor states represent the resting cognitive configurations of a dynamical cognitive system.
Attractor Basin	The region of cognitive state space surrounding a stable attractor within which perturbations are absorbed by the system's natural damping mechanisms and the system returns to equilibrium. The size of the attractor basin (its radius) is proportional to the epistemic damping capacity of the cognitive system.
Negative Attractor Basin	A stable but pathological cognitive configuration — such as chronic outrage, conspiratorial ideation, hypervigilance, or learned helplessness — that captures cognitive systems following high-amplitude perturbation away from healthy equilibria. Negative attractors are stable states and therefore resist ordinary corrective intervention once occupied.

Perturbation Field	The vector field of cognitive forces exerted by an algorithmic system across its user base, representing the direction and magnitude of cognitive displacement at each point in cognitive state space. The perturbation field is determined jointly by the AA value and the targeting precision of the delivery system.
Consensus Cognition	The shared cognitive baseline of a population — including common factual frameworks, causal reasoning norms, and intersubjective agreement about social reality — that enables collective decision-making, institutional function, and social coherence. Consensus cognition is the primary social asset the CSF aims to protect.
Generative Cognition	The healthy cognitive mode characterised by open curiosity, creative reasoning, adaptive belief updating, stable emotional regulation, and constructive social engagement. Generative cognition is the primary attractor state the CSF designates as the regulatory protection target.
Stability Band	The bounded region of algorithmic amplitude within which cognitive perturbation is insufficient to displace minds from their primary attractor basins. The stability band defines the regulatory safe zone and is bounded below by zero and above by the Amplitude Threshold (AT).
Amplitude Threshold (AT)	The critical value of algorithmic amplitude above which the probability of displacing a cognitive system from its primary attractor basin exceeds socially acceptable risk tolerances. AT is the principal regulatory parameter of the CSF and varies by population segment and exposure duration.
Cognitive State Vector (CSV)	A mathematical representation of an individual's cognitive configuration at a given moment, encoding belief coherence, emotional valence, epistemic openness, and social trust as four dimensional components. The CSV is the fundamental state variable of the CSF's mathematical model.
Population Drift	The aggregate directional shift of a population's cognitive state distribution over time, induced by sustained exposure to high-amplitude algorithmic content. Population drift is measured as the change in the centroid and dispersion of the distribution of CSVs across the population.
Resonance Cascade	A feedback condition in which algorithmically amplified content triggers self-reinforcing cognitive loops, accelerating displacement toward negative attractor basins. Resonance cascades occur when the perturbation field

aligns with the gradient of the potential energy landscape near a basin boundary, creating runaway displacement dynamics.

Epistemic Damping

The natural cognitive resistance of a well-functioning mind to displacement from its attractor basin, analogous to physical damping in oscillatory systems. Epistemic damping arises from metacognitive self-regulation, social feedback mechanisms, and stable belief infrastructure, and is quantified by the attractor basin radius r_P .

3. Scientific Foundations

3.1 Human Cognitive Stability

Human cognition may be formally modelled as a dynamical system — that is, a system whose state evolves over time according to rules that describe how the current state determines future states. This framing is not merely metaphorical. The qualitative behaviours exhibited by human cognitive systems — including stability under perturbation, sensitivity to initial conditions, bifurcations triggered by threshold-level stimuli, and convergence toward characteristic resting states — are precisely the hallmarks of nonlinear dynamical systems as studied in mathematics and physics. The cognitive state of an individual at any moment can be characterised by a vector in a high-dimensional state space, and the evolution of that state over time traces a trajectory through that space, attracted toward stable fixed points that correspond to characteristic cognitive configurations.

Healthy minds maintain cognitive stability through a suite of interlocking mechanisms. **Epistemic damping** refers to the capacity of a well-regulated cognitive system to absorb perturbations — incoming information, emotional stimuli, or social pressure — without being fundamentally displaced from its characteristic attractor state. This damping arises from metacognitive self-monitoring (the ability to observe and evaluate one's own reasoning processes), stable foundational beliefs that provide interpretive scaffolding for new information, and the regulatory influence of social networks that provide corrective feedback when individual cognition strays from shared norms. **Social feedback** is particularly important: human cognition is inherently social, and the corrective influence of trusted interlocutors constitutes a major source of stability for individual cognitive systems.

Metacognitive self-regulation — the capacity to identify and correct one's own epistemic errors — provides an additional layer of intrinsic stability that operates independently of social feedback.

The concept of **cognitive homeostasis** captures the aggregate effect of these mechanisms: a healthy cognitive system, like a healthy biological system, possesses a characteristic resting state toward which it tends to return after perturbation. This resting state is not rigid — healthy cognition accommodates new information, revises beliefs in response to evidence, and adapts to changing social environments. Rather, it is a dynamic equilibrium characterised by stable emotional regulation, coherent belief structure, appropriate epistemic openness, and functional social trust. The Cognitive Stability Framework takes cognitive homeostasis as its fundamental protective target, and algorithmic amplitude as the primary external force capable of disrupting it at scale.

3.2 Attractor Basins and Cognitive Architecture

The attractor basin concept provides the central architectural metaphor of the CSF's theoretical model. In dynamical systems theory, an attractor basin (or basin of attraction) is the set of all initial conditions from which a system's trajectory will converge to a given attractor. Geometrically, the attractor basin can be visualised using a **potential energy landscape** — a surface whose height at each point represents the energy cost of the system occupying that state. Attractors correspond to valleys in this landscape; perturbations correspond to forces that push the system up the valley walls; and epistemic damping corresponds to the restoring force that draws the system back toward the valley floor when displaced.

Within the CSF framework, three classes of attractor are distinguished. The **Primary Attractor Basin** corresponds to generative cognition — the healthy, open, adaptive cognitive mode characterised by stable emotional regulation, coherent beliefs, appropriate epistemic openness, and functional social trust. This basin is typically the deepest and widest attractor in the cognitive landscape of a well-functioning adult, reflecting the substantial epistemic damping capacity of a mature cognitive system with intact social connections. The **Secondary Attractor** represents a state of moderate belief rigidity — a cognitive configuration characterised by reduced epistemic openness and some degree of defensive ideation, but without the severe dysregulation associated with negative attractor states. Secondary attractors are less stable than the primary attractor but more accessible under conditions of moderate perturbation.

The **Negative Attractor Basins** represent the most concerning class of cognitive configurations: stable states characterised by chronic outrage, conspiratorial ideation, radicalized belief structures, hypervigilance, or learned helplessness. These basins are

distinguished by two critical properties. First, they are genuinely stable — once a cognitive system has been displaced into a negative attractor basin, the restoring forces toward the primary attractor are weak, and ordinary corrective interventions (factual corrections, social persuasion, counter-narratives) are largely ineffective. This stability arises because the negative attractor's cognitive architecture interprets disconfirming information as further evidence of its characteristic narrative (e.g., a conspiratorial framework that treats factual refutations as evidence of conspiracy). Second, negative attractor basins actively erode the epistemic infrastructure required for basin crossing — reducing social trust, narrowing epistemic openness, and destabilising belief coherence in ways that make escape progressively harder over time.

The energy landscape metaphor is productive precisely because it clarifies the role of amplitude in basin crossing. A cognitive system at rest in its primary attractor basin can be visualised as a ball resting in the deepest valley of the landscape. Perturbation forces — algorithmic content stimuli — push the ball up the valley walls. If the perturbation force is insufficient to push the ball over the basin wall, the ball returns to the valley floor when the perturbation ceases: this is epistemic damping. If the perturbation force exceeds the basin wall height — that is, if AA exceeds AT — the ball is pushed over the wall and may roll into an adjacent valley (a secondary or negative attractor basin). Critically, basin erosion progressively lowers the valley walls, so that the amplitude required to induce basin crossing decreases with sustained exposure. This is the fundamental dynamic that makes regulatory intervention both urgent and time-sensitive.

3.3 Algorithmic Perturbation Effects

Algorithmic content delivery systems exert perturbation forces on cognitive state vectors through four distinct and interacting mechanisms. Each mechanism contributes independently to total AA, and their interaction can produce non-additive perturbation effects under certain conditions.

(a) Amplitude Escalation refers to the progressive intensification of emotionally arousing content delivered by engagement-optimised systems. Because high emotional intensity drives measurable increases in engagement metrics (dwell time, shares, comments, and return visits), systems optimised for engagement exhibit a systematic bias toward escalating the affective intensity of delivered content over time. Amplitude escalation operates at both the individual session level — where content intensity increases as the user extends their session — and at the population level, where platform-wide engagement optimisation creates systematic upward pressure on the distribution of content intensity across the content ecosystem. The result is a secular increase in average $M(t)$ across the platform's content distribution that no individual content producer generates intentionally

but that the system's incentive structure reliably produces.

(b) Resonance Targeting refers to the capacity of precision-personalized systems to match delivered content to the specific proximity of a user's current cognitive state to particular attractor basins. A user whose CSV is near the boundary of a negative attractor basin can be identified through behavioural signals (content engagement patterns, session duration, interaction types), and content that is maximally resonant with that basin's characteristic narrative can be delivered at precisely the moment of maximum susceptibility. Resonance targeting is the mechanism by which algorithmic systems can efficiently convert moderate perturbation into basin crossing — concentrating perturbation force at the most strategically effective point in the individual's cognitive landscape rather than distributing it uniformly.

(c) Basin Erosion refers to the cumulative effect of repeated perturbation on the structural integrity of the primary attractor basin. Each episode of AA that approaches but does not quite cross the threshold AT exerts a partial but nonzero erosive effect on the basin walls — progressively shallowing the potential energy gradient that constitutes epistemic damping. This erosion is captured formally in the basin erosion model (Section 4.5), which shows that the basin radius r_P decays exponentially under sustained exposure to super-threshold amplitude. Basin erosion is the mechanism by which moderate, sustained algorithmic exposure that never triggers a single dramatic basin crossing event can nonetheless progressively undermine cognitive stability across a population.

(d) Cascade Induction refers to the triggering of self-reinforcing cognitive feedback loops — resonance cascades — in which algorithmic content stimulates the cognitive processes of the negative attractor basin (outrage, pattern-seeking, confirmation bias) which in turn increase engagement with further high-amplitude content, which further strengthens the basin's hold. Cascade induction is the mechanism responsible for the rapid, non-linear displacement events observed when a user's CSV crosses a basin boundary: the system's recommendation logic responds to the changed engagement pattern by delivering ever-more resonant content, accelerating the cascade.

Important Note: Cumulative Exposure Effects

Basin erosion and cascade dynamics mean that cognitive risk from algorithmic exposure is not linear with respect to time. A cognitive system exposed to moderate amplitude over a long period may be more vulnerable to basin crossing than a system exposed to high amplitude briefly. Regulatory frameworks that focus only on peak amplitude without accounting for cumulative exposure duration risk systematically underestimating the true cognitive risk posed by sustained moderate-

amplitude content delivery. The CSF's mathematical models explicitly account for cumulative exposure through the integral term in the basin erosion equation (Section 4.5).

Figure 1 below provides a visual representation of the cognitive attractor basin landscape, illustrating the energy landscape metaphor and the positions of the primary, secondary, and negative attractor basins relative to the amplitude threshold boundaries.

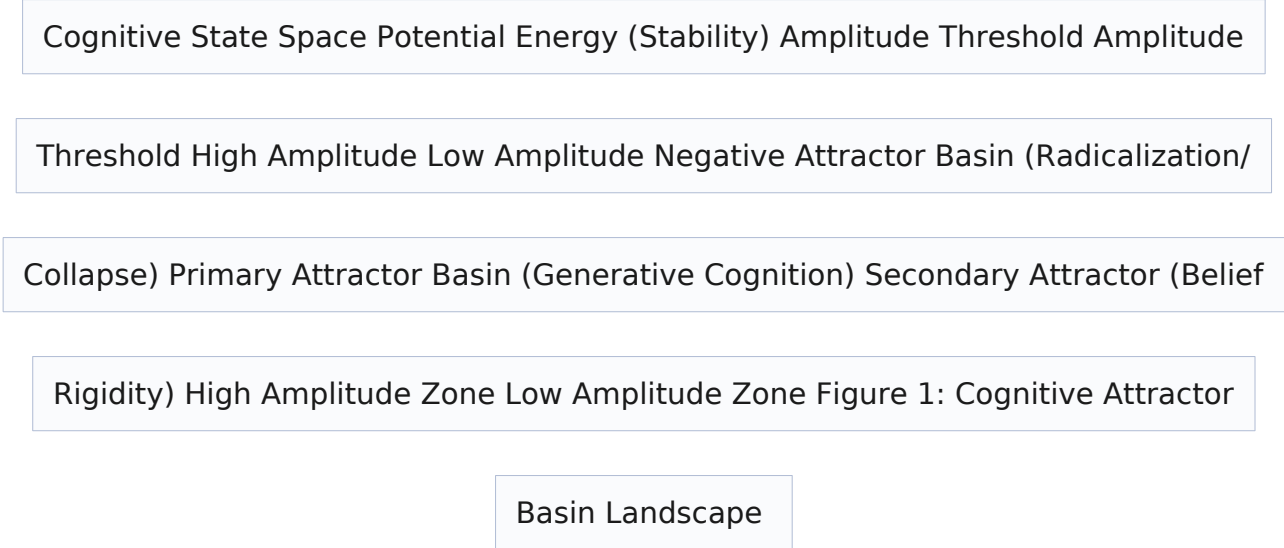


Figure 1: Cognitive Attractor Basin Landscape — potential energy representation of cognitive state space showing primary, secondary, and negative attractor basins. The ball indicates the current cognitive state resting in the primary basin. Dashed vertical lines represent Amplitude Threshold boundaries; colored shading indicates the extent of each basin.

4. Mathematical Models

This section presents the formal mathematical apparatus of the Cognitive Stability Framework. All equations are presented with full variable definitions and interpretive prose. The mathematical model is designed to be implementable with available data and to provide quantitative targets suitable for regulatory specification. Symbols used throughout are summarised in the Appendix Notation Table.

4.1 Algorithmic Amplitude Equation

The fundamental regulatory quantity of the CSF is **Algorithmic Amplitude (AA)**, defined

as the product of four components that together capture the full dimensionality of cognitively perturbative algorithmic delivery:

$$AA(t) = M(t) \times V(t) \times F(t) \times P(t)$$

This formulation encodes the empirically motivated claim that perturbation is multiplicatively determined by content intensity, delivery velocity, resonance precision, and targeting depth. Each component is defined and described below.

M(t) — Magnitude of content emotional/cognitive intensity (normalised 0–1): M(t) captures the intrinsic affective and cognitive arousal potential of content delivered at time t. It is computed from validated psycholinguistic intensity scoring applied to the delivered content, incorporating measures of emotional valence extremity, moral charge, threat salience, and epistemic destabilisation potential. A value of 0 corresponds to content with minimal arousal potential (factual, neutral, low-affect); a value of 1 corresponds to maximally intense content (extreme emotional charge, high moral urgency, direct threat activation). M(t) is arguably the most conceptually important component of AA, as it captures the intrinsic perturbation energy of individual content items.

V(t) — Velocity of content delivery (items per unit time): V(t) measures the rate at which content items are delivered to a user at time t, normalised against a reference delivery rate established empirically as the maximum that does not overwhelm epistemic processing capacity. High V(t) values prevent epistemic damping from completing its cycle between perturbation events, effectively compressing the recovery time available to the cognitive system and increasing the cumulative perturbation load per unit time. Velocity is perhaps the most directly regulable component of AA, as it is controlled by platform-side delivery throttling mechanisms.

F(t) — Frequency resonance factor (0–1): F(t) captures the degree of alignment between the type and framing of delivered content and the user's current attractor proximity, as inferred from behavioural engagement signals. A high F(t) value indicates that delivered content is highly resonant with the user's current cognitive state — that is, it targets themes, frames, and narratives that are maximally engaging given the user's current position in cognitive state space. F(t) is the formal representation of the resonance targeting mechanism described in Section 3.3(b), and it is the component of AA most directly influenced by personalisation algorithm design.

P(t) — Targeting precision coefficient (0–1): P(t) measures the depth of personalisation applied to the content delivery decision at time t — specifically, the number and resolution of personalisation dimensions active in the recommendation system. A P(t) of 0 corresponds to fully undifferentiated delivery (identical content to all users); a P(t) of 1

corresponds to maximally precise individual targeting based on the full available behavioural and inferred psychological profile. $P(t)$ modulates the efficiency with which the other AA components are applied — high precision ensures that M , V , and F are concentrated on the individuals and moments where they will produce maximum perturbation effect.

Key Property: Dimensionless AA

Because all four components of AA are normalised (M , F , P to $[0,1]$ and V to a normalised ratio), the resulting AA value is dimensionless, ranging from 0 (no perturbation) to 1 (maximum theoretical perturbation). This dimensionless property enables direct cross-platform comparison of amplitude values, independent of platform size, content category, or delivery architecture. It is the critical property that makes AA usable as a universal regulatory standard.

4.2 Cognitive State Vector

The cognitive configuration of an individual at time t is represented as a four-dimensional **Cognitive State Vector (CSV)**:

$$CSV(t) = [B(t), E(t), O(t), T(t)]$$

Each component captures a distinct and empirically measurable dimension of cognitive state.

B(t) — Belief Coherence $[0, 1]$: $B(t)$ measures the internal consistency and structural integrity of an individual's belief system at time t . A value of 1 indicates fully coherent, internally consistent beliefs with clear logical relationships between component beliefs. A value approaching 0 indicates fragmented, contradictory, or incoherent beliefs. High-amplitude content delivery reduces $B(t)$ by introducing rapidly cycling narratives that are difficult to integrate coherently, forcing the belief system into a state of chronic provisional revision that degrades overall coherence.

E(t) — Emotional Valence $[-1, +1]$: $E(t)$ captures the net affective charge of the individual's emotional state at time t . Positive values indicate a net positive affective charge (calm, engaged, optimistic); values near zero indicate emotional neutrality; negative values indicate net negative affect (anxiety, anger, despair). High-amplitude content delivery systematically drives $E(t)$ toward negative values through preferential delivery of threat-relevant, outrage-inducing, or anxiety-provoking content, as these affective states are associated with the highest engagement metrics.

O(t) — Epistemic Openness [0, 1]: O(t) measures the individual's capacity for belief revision in response to new evidence at time t. A value of 1 indicates full epistemic openness — willingness to update beliefs in response to disconfirming evidence. A value approaching 0 indicates epistemic closure — systematic rejection of disconfirming information, characteristic of strongly fortified negative attractor states. High-amplitude resonance targeting systematically reduces O(t) by reinforcing existing belief structures rather than exposing users to epistemically diverse content.

T(t) — Social Trust [0, 1]: T(t) measures the individual's confidence in shared social reality — trust in institutions, media, interpersonal relationships, and collective epistemic authorities at time t. T(t) is the most socially consequential component of the CSV, as its degradation directly impairs the social feedback mechanisms that constitute a major source of epistemic damping. High-amplitude content delivery systematically reduces T(t) through sustained exposure to narratives emphasising institutional corruption, elite manipulation, and the unreliability of shared information sources.

Definition: Healthy CSV Configuration

A CSV configuration consistent with the primary attractor basin (Generative Cognition) is characterised by: $B(t) > 0.70$, $E(t) > -0.20$, $O(t) > 0.55$, $T(t) > 0.60$. Departure from these ranges in any component increases the probability of transition toward secondary or negative attractor states.

4.3 Attractor Basin Definitions

The two principal attractor basins relevant to regulatory analysis are formally defined as follows.

The **Primary Attractor Basin (Ω_P)** is the set of all Cognitive State Vectors within a characteristic distance of the primary attractor centroid CSV_{P*}:

$$\Omega_P = \{ \text{CSV} : ||\text{CSV} - \text{CSV}_{P^*}|| < r_P \}$$

where CSV_{P*} = [0.80, 0.25, 0.70, 0.75] is the primary attractor centroid (an idealized healthy cognitive configuration), $||\cdot||$ denotes the Euclidean norm in the four-dimensional cognitive state space, and r_P is the basin radius — the **epistemic damping constant** that quantifies the magnitude of perturbation the cognitive system can absorb without leaving the primary attractor basin. A larger r_P indicates a more robust, well-damped cognitive system; a smaller r_P indicates a more vulnerable system closer to basin boundaries.

The **Negative Attractor Basin (Ω_N)** is defined analogously:

$$\Omega_N = \{ CSV : ||CSV - CSV_{N^*}|| < r_N \}$$

where $CSV_{N^*} = [0.25, -0.70, 0.15, 0.20]$ is the negative attractor centroid (a pathological cognitive configuration characterised by low belief coherence, strongly negative emotional valence, epistemic closure, and severely depleted social trust), and r_N is the negative basin radius. Notably, r_N is typically smaller than r_P for healthy populations but grows as population-level basin erosion progresses — reflecting the self-reinforcing stability of pathological cognitive states once established.

4.4 Stability Threshold Equations

The **Amplitude Threshold (AT)** is defined as the critical amplitude value beyond which the probability of an individual being displaced from the primary attractor basin within a defined exposure window exceeds a socially acceptable risk tolerance ε :

$$P(\text{displacement} \mid AA > AT) > \varepsilon$$

where ε is the regulatory displacement risk tolerance, proposed at $\varepsilon = 0.05$ (a 5% per-exposure probability of basin displacement for a typical individual). This value is calibrated to be analogous to standard safety tolerances used in public health regulation (e.g., acceptable individual risk thresholds in environmental exposure standards).

The **Stability Condition** — the fundamental regulatory requirement of the CSF — is:

$$AA(t) < AT \quad \text{for all } t \in [0, T]$$

This condition states that algorithmic amplitude must remain below the amplitude threshold throughout all periods of platform operation. Regulatory compliance is defined as satisfaction of the Stability Condition across all rolling 30-day windows, subject to the tiered threshold specifications defined in Section 8.1.

4.5 Negative Attractor Formation Model

Basin erosion under sustained high-amplitude exposure is modelled by the following differential decay equation for the primary attractor basin radius r_P :

$$r_P(t) = r_P(0) \times \exp\left(-\lambda \times \int_0^t [AA(\tau) - AT]_+ d\tau\right)$$

where:

- **$r_P(0)$** is the initial basin radius, representing the healthy epistemic damping

capacity of the individual or population segment before algorithmic exposure;

- λ is the erosion rate constant, an empirically determined parameter that captures the rate at which excess amplitude degrades epistemic damping capacity. λ varies by population segment — higher for minors (less developed epistemic infrastructure), lower for populations with strong social support networks;
- $[AA(\tau) - AT]_+ = \max(AA(\tau) - AT, 0)$ is the positive-part operator, which ensures that only amplitude in excess of the threshold contributes to basin erosion. Amplitude below AT does not erode the basin; amplitude above AT erodes it at a rate proportional to the excess.

The exponential form of this equation has an important regulatory implication: basin erosion is irreversible in the short term and accelerating over time. As $r_P(t)$ decreases, the effective threshold AT decreases proportionally (since the system requires less amplitude to cross a shallower basin wall), creating a positive feedback dynamic in which erosion begets further susceptibility to erosion. This is the mathematical formalization of the cumulative exposure concern noted in Section 3.3. Early regulatory intervention — before significant erosion has accumulated — is dramatically more effective than late intervention, because the integral in the exponent grows monotonically with continued super-threshold exposure.

Critical Regulatory Implication

The basin erosion model demonstrates mathematically that regulatory inaction compounds harm non-linearly. A population exposed to $AA = 0.75$ (above $AT_{\text{general}} = 0.60$) for 12 months will exhibit substantially lower residual r_P than one that received regulatory intervention at month 3, even if AA returns to compliant levels at month 12. This argues strongly for early-intervention regulatory postures and against "wait and see" approaches to amplitude governance.

4.6 Population-Level Stability Equation

Individual-level basin dynamics aggregate to a population-level measure through the **Population Stability Index (PSI)**:

$$PSI(t) = (1/N) \times \sum_{i \in I} \mathbb{I}[CSV_i(t) \in \Omega_P]$$

where N is the total population size and $\mathbb{I}[\cdot]$ is the indicator function that takes the value 1 if individual i 's Cognitive State Vector lies within the primary attractor basin Ω_P at time t , and 0 otherwise. $PSI(t)$ is therefore simply the proportion of the population currently residing within the primary attractor basin — the fraction of individuals in a generative, cognitively stable state.

The **Regulatory PSI Target** is:

$$\text{PSI}(t) \geq 0.85 \quad \text{for all } t$$

This target — that at least 85% of the platform-exposed population must remain within the primary attractor basin at all times — is calibrated to maintain the epistemic commons at a level sufficient for reliable democratic deliberation and collective decision-making. As discussed in Section 7.2, PSI values below 0.85 correspond to measurable strain on cooperative institutions, with progressively more severe social consequences as PSI declines further.

Figure 2 below provides a schematic overview of the causal chain from algorithmic content delivery through the mathematical model to population stability outcomes.

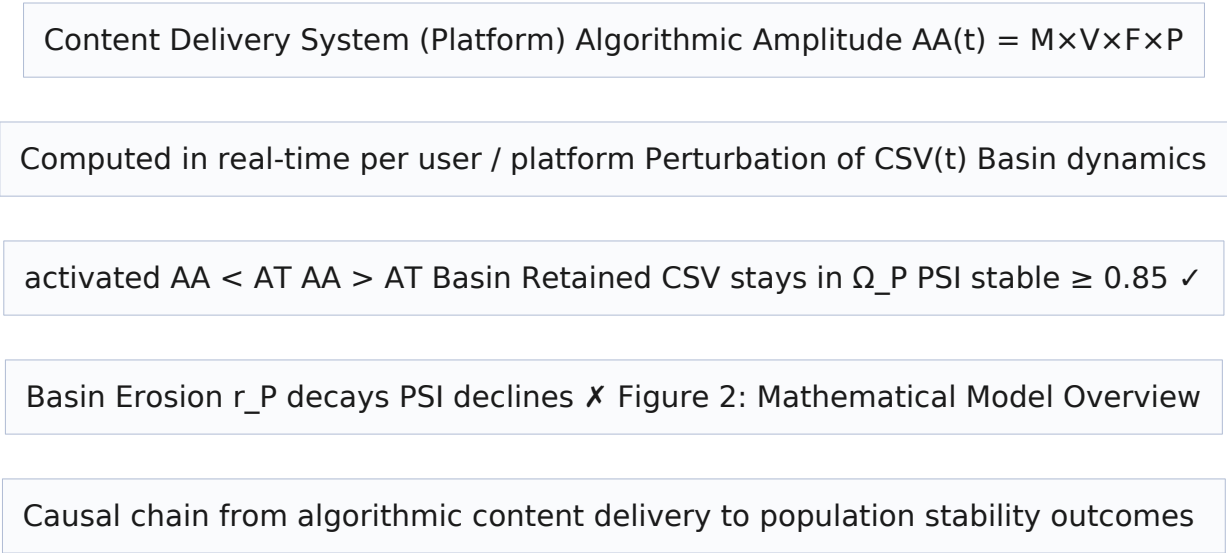


Figure 2: Mathematical Model Overview — causal chain from algorithmic content delivery to population stability outcomes. The green pathway represents compliant operation (AA below threshold); the red pathway represents non-compliant operation triggering basin erosion and PSI decline.

5. Visual Diagrams

The following five figures provide visual representations of key dynamics and structures within the Cognitive Stability Framework. Each figure is a standalone analytical contribution and is referenced by number throughout the remainder of the document.

Figure 3: Algorithmic Amplitude Dynamics

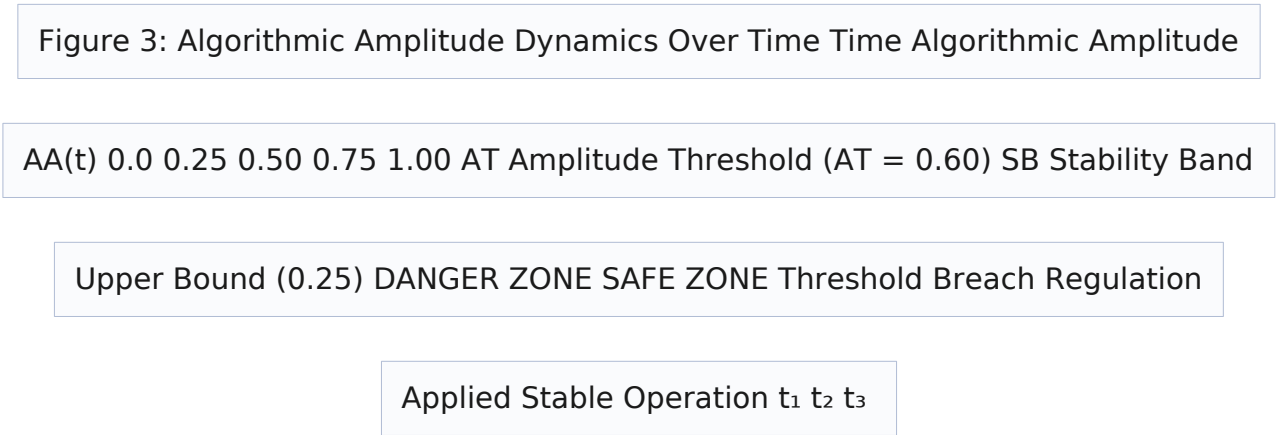


Figure 3: Algorithmic Amplitude Dynamics over time showing threshold breach at t_1 , regulatory intervention at t_2 , and return to safe operating range toward t_3 . The shaded red region indicates the Danger Zone ($AA > AT$); the shaded green region indicates the Safe Zone ($AA < AT$) following regulation.

Figure 4: Cognitive Attractor Basin Contour Map

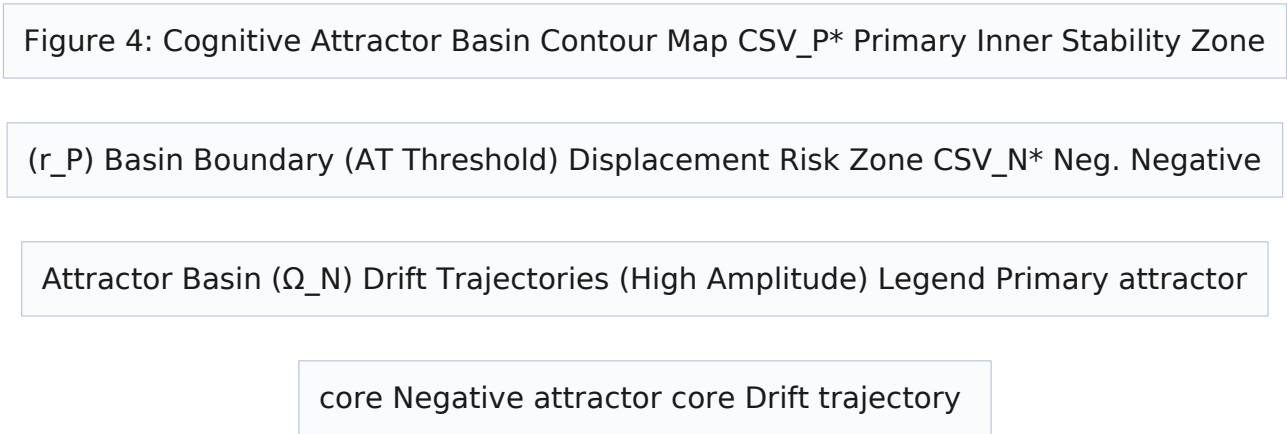
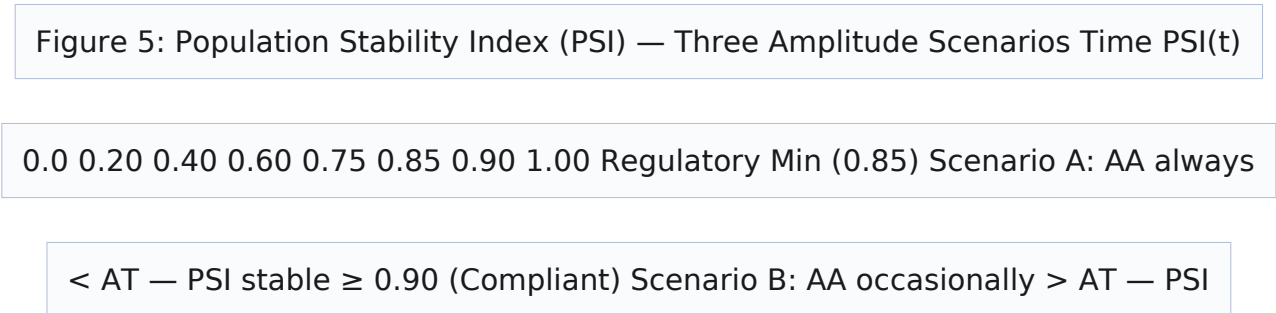


Figure 4: Cognitive Attractor Basin contour map showing primary attractor centroid (CSV_P*), inner stability zone (r_P radius), basin boundary at amplitude threshold, displacement risk zone, and the negative attractor basin (Ω_N) with representative drift trajectories under high-amplitude conditions.

Figure 5: Population Stability Index — Three Scenarios



oscillates (Borderline) Scenario C: AA chronically > AT — PSI declines to 0.40 (Non-

Compliant) A B C

Figure 5: Population Stability Index (PSI) trajectories under three amplitude scenarios — Scenario A: compliant operation (AA consistently below AT), Scenario B: borderline operation (occasional threshold breaches), and Scenario C: non-compliant operation (chronic threshold breaches). The dashed orange horizontal line marks the regulatory minimum PSI of 0.85.

Figure 6: Perturbation Vector Field

Figure 6: Perturbation Vector Field Across Cognitive State Space Belief Coherence

B(t) Epistemic Openness O(t) 0.0 0.25 0.50 0.75 1.00 0.0 0.25 0.50 0.75 1.00

CSV_P* Stable Region (Inward-pointing field) Perturbation Dominant Zone

Perturbation Dominant Zone

Figure 6: Perturbation Vector Field across cognitive state space (Belief Coherence vs. Epistemic Openness). Navy arrows near the centre indicate field forces directed toward the primary attractor (CSV_P*). Red arrows at the periphery indicate large outward-directed perturbation forces characteristic of high-amplitude conditions, pointing away from the stable region toward displacement.

Figure 7: Population-Level Cognitive State Distribution Drift

Figure 7: Population-Level Cognitive State Distribution Drift Cognitive State —

Generative (right) to Destabilized (left) Population Density ← Destabilized

Generative → Primary Basin Boundary Negative Basin Boundary Legend t = 0

(Initial, narrow) t = T/2 (Shifted, wider) t = T (Far left, dispersed) Population Drift

(Leftward under sustained AA)

Figure 7: Population-level cognitive state distribution drift under sustained high-amplitude exposure. The three overlapping distributions represent the population at $t = 0$ (initial, navy), $t = T/2$ (intermediate, blue-gray, dashed), and $t = T$ (final, red, dashed). Progressive leftward drift and distribution broadening indicate increasing proportions of the population entering destabilized cognitive states beyond the negative basin boundary.

6. Risk Analysis

6.1 Effects of High Amplitude on Consensus Cognition

Consensus cognition — the shared epistemic commons upon which collective decision-making, institutional legitimacy, and social coherence depend — is the first and most fundamental casualty of sustained high-amplitude algorithmic content delivery. The mechanism is straightforward: high-amplitude content delivery systems, optimised for individual engagement, systematically prioritise content that resonates with individual cognitive profiles rather than content that reinforces shared factual frameworks. Over time, this creates diverging information environments across the population — not because individuals have different access to information, but because algorithmic personalisation ensures that they receive systematically different selections from the same underlying information ecosystem. The aggregate effect is the progressive erosion of the shared cognitive substrate that makes intersubjective agreement possible.

This erosion proceeds through three identifiable stages. **Stage 1 — Epistemic Fragmentation** is characterised by the emergence of divergent factual beliefs across population sub-groups — different understandings of basic empirical facts about health, economics, history, and current events. At this stage, intergroup disagreement is primarily factual: groups hold different beliefs about what is true. Epistemic fragmentation is the earliest detectable stage of consensus cognition erosion and the most amenable to intervention, as shared factual frameworks can in principle be restored through common information environments.

Stage 2 — Narrative Tribalism is characterised by the development of group-specific reality tunnels — self-contained interpretive frameworks that organise factual claims, causal attributions, and moral evaluations into coherent but mutually incompatible worldviews. At this stage, disagreement is no longer merely factual but structural: different groups are not simply disputing facts, but applying fundamentally different frameworks for interpreting evidence, assigning causation, and evaluating claims. Stage 2 is significantly

harder to reverse than Stage 1, because counter-narrative information is typically interpreted through the existing narrative framework rather than treated as epistemically independent evidence.

Stage 3 — Coherence Collapse represents the terminal stage of consensus cognition degradation, characterised by the inability to form reliable shared decisions even within groups that nominally share a narrative framework. At this stage, even intragroup epistemic trust has been eroded — the capacity for coordinated collective reasoning has broken down at a sufficiently fine-grained level that collective action becomes unreliable. Coherence collapse is the point at which the social consequences of epistemic fragmentation become irreversible through ordinary epistemic means: the shared cognitive infrastructure required to negotiate agreement has been destroyed.

6.2 Social Stability Implications

The connection between cognitive destabilization and social instability is not correlational but mechanistic. Political polarization increases directly as a function of epistemic fragmentation: when groups hold irreconcilable factual beliefs, political disagreements lose their negotiable character and become identity conflicts that admit no rational resolution. Rising rates of conspiracy belief are a predictable consequence of Stage 2 narrative tribalism, as conspiratorial frameworks provide coherent causal accounts within self-contained reality tunnels that are impervious to external factual correction. Declining institutional trust is both a symptom and an amplifier of consensus cognition erosion: when shared epistemic authorities (scientific institutions, journalistic organisations, governmental bodies) are systematically discredited by high-amplitude algorithmic content, the social mechanisms that enable epistemic repair are themselves disabled.

Increased intergroup hostility follows predictably from the combination of epistemic fragmentation and narrative tribalism. When outgroup members are perceived not merely as holding different opinions but as inhabiting categorically different realities, the affective and motivational basis for prosocial intergroup behaviour is severely attenuated. The breakdown of cooperative norms — reciprocity, trust in shared agreements, willingness to accept collective decisions — is the downstream consequence of this hostility amplified to social scale. It is critical to understand that these outcomes are not incidental by-products of algorithmic content systems; they are the foreseeable and predictable social consequences of engagement-optimised amplitude delivery at scale, operating through the mechanisms described in Sections 3 and 4.

6.3 Long-Term Systemic Risks

Beyond the immediate social consequences of cognitive destabilization, the CSF identifies

four categories of systemic risk that threaten institutional and societal stability over longer time horizons.

1. Democratic Integrity Risk. Democratic institutions presuppose a population capable of collective rational decision-making — a minimal level of shared factual understanding, functional epistemic trust, and cooperative norm maintenance sufficient to make majority decision-making meaningful rather than merely a mechanism for resolving factional conflicts. Population-level consensus cognition erosion directly undermines these presuppositions. When PSI falls below critical thresholds, democratic deliberation degenerates into epistemic warfare in which factual claims function primarily as rhetorical weapons rather than jointly evaluated evidence. The long-term integrity of democratic institutions depends upon the maintenance of a cognitive commons that high-amplitude algorithmic systems systematically erode.

2. Public Health Risk. Cognitive destabilization correlates with measurable mental health deterioration at population scale. The emotional valence dimension of the CSV ($E(t)$) captures the most direct pathway: sustained negative emotional valence associated with high-amplitude content delivery (anxiety, outrage, fear, despair) is a well-established risk factor for depression, anxiety disorders, and related conditions. Beyond direct affective effects, the social isolation associated with narrative tribalism (reduction in bridging social capital, retreat into homogeneous epistemic communities) is an independent risk factor for multiple adverse health outcomes. At population scale, these effects constitute a significant and systematically underestimated public health burden attributable to unconstrained algorithmic amplitude.

3. Economic Stability Risk. Market economies depend upon epistemic coordination — shared understanding of property rights, contract enforceability, price information reliability, and regulatory legitimacy — as the foundation for efficient economic exchange. Epistemic fragmentation reduces this coordination capacity, increasing transaction costs as parties must verify information that would otherwise be assumed shared, generating market irrationality as investment decisions are increasingly informed by narrative-tribal rather than evidence-based causal models, and undermining regulatory legitimacy in ways that create systematic governance gaps. These effects aggregate into measurable reductions in economic efficiency and coordination capacity at the national and international level.

4. Security Risk. Negative attractor basins — particularly those characterised by radicalized belief structures, conspiratorial ideation, and intergroup hostility — constitute direct security vectors. Individuals and groups occupying negative attractor basins are significantly more susceptible to recruitment into organized social violence, more likely to perceive violence as epistemically justified within their narrative framework, and more

resistant to de-radicalisation interventions that depend upon shared epistemic norms. At population scale, the growth of negative attractor basin occupancy represents a measurable and growing security risk that conventional security frameworks are poorly equipped to address, because the root cause is not individual pathology but systemic algorithmic perturbation.

7. Regulatory Rationale

7.1 Why Regulation Is Necessary

The case for regulatory intervention in algorithmic amplitude rests upon a foundational observation in welfare economics: market mechanisms generate efficient outcomes only when the costs of production are fully borne by the producers, and the benefits are fully received by the consumers. When significant costs are displaced onto parties who are not participants in the market transaction — what economists term externalities — market outcomes are systematically inefficient, and regulation is the standard corrective mechanism. Algorithmic amplitude presents a textbook case of negative externality at extraordinary scale. The costs of cognitive destabilization — mental health deterioration, social fragmentation, institutional erosion, democratic degradation — are borne overwhelmingly by the affected populations and by society as a whole, not by the platform operators whose engagement-optimised amplitude delivery produces them. Platform operators, on the contrary, capture the revenues generated by high-amplitude engagement. This cost-benefit asymmetry creates a systematic and irresolvable market incentive toward high-amplitude operation that no individual market actor has incentive to unilaterally correct.

The engagement-amplitude dynamic is particularly resistant to market self-correction because the consumer-facing product (content delivery) conceals its cognitively perturbative properties from users. Unlike environmental pollutants, which may be detected and attributed, algorithmic amplitude operates through mechanisms — progressive resonance targeting, basin erosion, cascade induction — that are invisible to the individual user and attributable to no single content item or delivery event. This information asymmetry between platform operators (who have full visibility into their AA metrics and their relationship to engagement outcomes) and users (who experience only the aggregate cognitive effects without access to the causal mechanisms) is a second

independent justification for regulatory intervention: market failures arising from information asymmetry are precisely the domain in which mandatory disclosure and performance standards are the appropriate policy tool.

Furthermore, network effects and market concentration in the platform economy mean that individual users cannot effectively exit high-amplitude platforms without incurring significant social costs — loss of communication channels, professional networks, and community connections. The absence of meaningful consumer exit options eliminates the competitive pressure that would otherwise discipline amplitude maximization behaviour. Regulatory intervention is therefore not merely justified but necessary as the only available mechanism for correcting the amplitude market failure.

7.2 How Amplitude Affects Societal Coherence

The relationship between PSI and societal coherence is not merely correlational but structurally determined by the role of shared cognition in enabling collective institutions. The CSF proposes three critical PSI thresholds with qualitatively distinct societal consequences. When **PSI falls below 0.85**, cooperative institutions begin to show measurable strain: political deliberation becomes more contentious, institutional trust declines at measurable rates, and the costs of intergroup coordination increase. At this threshold, the epistemic commons remains structurally intact but its load-bearing capacity is compromised — it can sustain ordinary institutional function but is vulnerable to acute destabilization events (crises, elections, public health emergencies) that require high levels of collective rational coordination.

When **PSI falls below 0.70**, democratic processes become systematically unreliable. Electoral outcomes increasingly reflect narrative-tribal identity rather than evidenced policy evaluation; legislative deliberation loses its epistemic character and becomes primarily a mechanism for resolving factional power conflicts; judicial authority is increasingly contested by groups that reject shared epistemic norms. At this threshold, the infrastructure of democratic decision-making remains formally intact but is functionally compromised in ways that render collective decisions increasingly unrepresentative of genuine informed preferences.

When **PSI falls below 0.55**, social contract mechanisms begin to fail. The mutual obligations and shared norms that underpin reciprocal social arrangements — law compliance, tax remittance, institutional deference, intergroup cooperation — lose their motivational basis for growing segments of the population. At this threshold, the minimum epistemic consensus required for stable social order can no longer be maintained through ordinary institutional means. Amplitude regulation is therefore not merely a matter of individual wellbeing or platform governance — it is a prerequisite for the continued

functioning of the institutional architecture of modern democratic societies.

7.3 Benefits of Establishing Safe Operating Ranges

The regulatory benefits of establishing and enforcing safe operating ranges for algorithmic amplitude extend across multiple dimensions. At the societal level, amplitude regulation preserves the epistemic commons — the shared cognitive substrate of shared factual understanding, intersubjective trust, and cooperative norm maintenance that enables collective decision-making, institutional legitimacy, and social coherence. This is perhaps the most fundamental societal good that CSF regulation protects, and its value is most clearly perceived in its absence: epistemic commons restoration after advanced consensus cognition erosion is extremely costly and may be functionally impossible within relevant political time horizons.

At the institutional level, amplitude regulation protects the integrity of democratic deliberation by ensuring that the population retains the shared cognitive infrastructure required for meaningful collective decision-making. It reduces the public mental health burden associated with sustained high-amplitude exposure — a burden currently externalised onto health systems that bear the costs of anxiety, depression, and social isolation at population scale without receiving any portion of the revenues generated by the amplitude delivery that produces these outcomes. At the platform level, early voluntary adoption of amplitude standards carries significant strategic advantages: platforms that establish credible amplitude governance ahead of mandatory requirements will have first-mover advantages in regulatory negotiation, will face lower compliance transition costs than those required to retrofit amplitude monitoring infrastructure under regulatory deadline pressure, and will benefit from the reputational advantages of demonstrable commitment to cognitive safety standards that increasingly sophisticated users and advertisers will demand. The long-term liability exposure of platforms that deliberately or negligently maintain super-threshold amplitude delivery is substantial and growing as the regulatory framework for algorithmic accountability matures.

8. Safe Operating Range Specification

8.1 Amplitude Threshold Definitions

The CSF establishes a three-tier regulatory zone structure based on empirically motivated amplitude threshold values calibrated to the epistemic damping capacity of the general adult population. These tiers provide a graduated regulatory response that permits proportionate oversight without imposing unnecessary compliance burdens on low-amplitude platforms.

Zone	AA Range	Characterisation	Regulatory Status
Green Zone	$AA \leq 0.25$	Minimal perturbation; within natural epistemic damping capacity.	No restriction required. Standard platform operation permitted without amplitude-specific oversight.
Amber Zone	$0.25 < AA \leq 0.60$	Moderate perturbation; approaching epistemic damping capacity limits.	Active monitoring required. Platforms must report quarterly PSI metrics and computed AA values to ACIA.
Red Zone	$AA > 0.60$	Excessive perturbation; exceeds epistemic damping capacity for general population.	Immediate regulatory review required. Mandatory amplitude reduction protocols must be activated within 72 hours. Sustained Red Zone operation (more than 72 continuous hours) triggers formal enforcement action.

Calibration Basis

The threshold values $AT = 0.25$ (Amber boundary) and $AT = 0.60$ (Red boundary) are calibrated to the estimated epistemic damping capacity of a healthy adult population with functional social networks and intact epistemic infrastructure. $AT = 0.25$ represents the approximate amplitude level at which measurable but sub-clinical perturbation effects begin to emerge in controlled exposure studies; $AT = 0.60$ represents the estimated amplitude level at which the probability of basin displacement in a single exposure window exceeds the regulatory tolerance of $\epsilon = 0.05$. Both values are subject to revision as empirical calibration studies are conducted by the CSSAP.

8.2 Cognitive Tolerance Limits

The general-population amplitude thresholds defined in Section 8.1 are not uniformly applicable across all population segments. The basin radius r_P — and therefore the effective amplitude threshold AT — varies systematically with several characteristics of the target population. The CSF mandates differential amplitude thresholds for three protected population segments, each reflecting empirically grounded differences in epistemic damping capacity.

Minors (under 18): The prefrontal cortex — the neural substrate of metacognitive self-regulation and the primary source of the top-down epistemic damping that protects adult cognitive systems from basin displacement — remains in active development until the mid-twenties. This developmental incompleteness means that minors have substantially smaller effective basin radii than adults, and are therefore significantly more susceptible to amplitude-induced basin crossing at any given AA level. The CSF proposes a mandatory threshold of **$AT_{\text{minor}} = 0.15$** — substantially below the Green Zone upper boundary for adults — reflecting the heightened vulnerability of developing cognitive systems. Platforms with significant minor user populations must apply minor-specific amplitude constraints to all content delivered to users identified as or likely to be minors.

General Population: The standard threshold **$AT_{\text{general}} = 0.60$** applies to adult users without identified vulnerability factors. This is the primary regulatory parameter for the majority of platform interactions and corresponds to the Red Zone activation threshold defined in Section 8.1.

Vulnerable Groups: Individuals who are near basin boundaries due to pre-existing conditions (anxiety disorders, depression, active grief, recent social isolation, identified radicalization risk) have reduced residual epistemic damping capacity — their effective r_P is smaller than the population average, meaning that a given AA value carries higher displacement risk for these individuals than for the general population. The CSF proposes a protective threshold of **$AT_{\text{vulnerable}} = 0.30$** for content delivered to users flagged as belonging to vulnerable segments. Platforms must develop and validate vulnerability identification protocols subject to CSSAP review.

Cumulative exposure duration is a further modifier of effective tolerance. The basin erosion model (Section 4.5) implies that effective AT decreases as a function of cumulative super-threshold exposure. Platforms must therefore maintain longitudinal exposure tracking to identify users whose cumulative exposure history places them in a lower effective tolerance category, and adjust delivered amplitude accordingly.

8.3 Recommended Regulatory Targets

The CSF establishes the following formal regulatory targets as mandatory compliance requirements for all covered platforms (platforms with ≥ 1 million monthly active users in any jurisdiction that has adopted the CSF regulatory framework):

1. **Population Stability Index:** $PSI(t) \geq 0.85$ must be maintained for all t in any rolling 30-day window across the platform-exposed population. PSI computation must be performed using validated methodology approved by the CSSAP and submitted to ACIA with quarterly reporting.
2. **Amplitude Ceiling:** $AA(t) < AT_{\text{general}} = 0.60$ must be satisfied for all t in any rolling 30-day window across the general adult population segment. Breach of this target triggers Amber Zone or Red Zone oversight requirements as specified in Section 8.1.
3. **Minor Protection Ceiling:** $AA(t) < AT_{\text{minor}} = 0.15$ must be satisfied for all content delivered to identified or likely minor users, with no rolling window exception — minor amplitude limits are point-in-time requirements, not rolling averages.
4. **Quarterly Reporting:** All covered platforms must submit quarterly reports to ACIA including computed AA metrics (disaggregated by content category, user demographic segment, and geographic region), PSI estimates, and documentation of any threshold breaches and corrective actions taken.
5. **Independent Audit:** Annual third-party audits of platform amplitude computation methodology must be conducted by CSSAP-approved auditors and submitted to ACIA with audit reports publicly disclosed (with commercially sensitive details appropriately redacted).

9. Governance Structure

9.1 Independent Oversight Body

The CSF proposes the establishment of an **Algorithmic Cognitive Impact Authority (ACIA)** as the primary regulatory body responsible for amplitude governance. ACIA is envisioned as an independent statutory authority — structurally analogous to existing independent financial regulators — with the legal independence, technical capacity, and enforcement powers required to effectively regulate algorithmic amplitude across the platform ecosystem. ACIA's operational independence from both government and industry

is a non-negotiable design requirement: the cognitive stability risks addressed by the CSF are sufficiently fundamental to democratic function that regulatory capture by either governmental or commercial interests would undermine the Authority's legitimacy and effectiveness.

ACIA's statutory powers include: receipt and review of mandatory platform amplitude reports; authority to conduct independent amplitude audits (including access to platform algorithmic systems, content delivery logs, and personalization infrastructure under appropriate confidentiality protections); issuance of formal compliance notices and compliance rectification orders; imposition of mandatory amplitude reduction orders with specified timelines and verification requirements; and authority to levy financial penalties for sustained Red Zone operation, non-reporting, and audit obstruction. Penalties are proposed to scale with platform revenue (a percentage of global annual revenue analogous to GDPR penalty structures) to ensure proportionality across platforms of differing scale.

ACIA membership is designed to ensure multi-disciplinary expertise and civil society representation. The proposed composition includes: cognitive scientists specialising in attractor dynamics and epistemic systems; behavioural economists with expertise in incentive structures and externality analysis; data scientists with platform algorithmic expertise; legal scholars specialising in technology regulation and constitutional constraints on content-related regulation; civil society representatives drawn from diverse cultural and demographic backgrounds; and platform technology specialists appointed in a non-voting advisory capacity to provide technical context for regulatory decisions. Members serve staggered 5-year terms; conflict-of-interest disclosure and cooling-off periods apply to all appointments.

9.2 Scientific Advisory Panel

The **Cognitive Stability Scientific Advisory Panel (CSSAP)** is proposed as a permanent, independent scientific body operating in parallel with ACIA and providing the empirical foundation for ACIA's regulatory decisions. CSSAP's role is strictly scientific: it does not exercise regulatory authority but provides authoritative scientific guidance on amplitude threshold calibration, measurement methodology validation, emerging risk identification, and state-of-knowledge synthesis. The separation of scientific advisory and regulatory enforcement functions is a deliberate design choice intended to preserve the integrity of both functions — ensuring that regulatory decisions are science-informed without placing the burden of enforcement decisions on scientists, and that scientific findings are not distorted by regulatory or political pressures.

CSSAP's specific responsibilities include: annual review and updating of amplitude threshold values (AT_general, AT_minor, AT_vulnerable) as measurement science and empirical

calibration studies mature; review and validation of platform-submitted AA computation methodology to ensure that platforms are computing amplitude consistently with CSSAP-approved standards; peer-reviewed publication of an annual State of Cognitive Stability report synthesising available evidence on population-level cognitive dynamics and algorithmic amplitude effects; and proactive identification of emerging amplitude risk vectors — including novel content delivery mechanisms, new forms of resonance targeting, or platform architecture changes that may circumvent existing amplitude definitions.

CSSAP membership consists of rotating academic appointments drawn from cognitive science, computational social science, information theory, and related disciplines, with 3-year staggered terms to ensure continuity of institutional knowledge. All CSSAP members are required to make full conflict-of-interest disclosures (including research funding sources, consultancy relationships, and equity holdings) and are prohibited from accepting platform-funded research during their term of appointment. CSSAP deliberations and technical reports are published in full; dissenting views from panel members are recorded and made public.

9.3 Transparency and Reporting Requirements

Transparency is a foundational principle of the CSF governance architecture. The failure to disclose amplitude metrics and their cognitive consequences to affected populations is a form of structural information asymmetry that undermines both individual autonomy and democratic accountability. The following reporting requirements are mandated for all covered platforms:

- **Quarterly AA Reports:** Platforms must publish quarterly reports to ACIA providing computed AA metrics disaggregated by content category (news, entertainment, political content, health information, etc.), user demographic segment (age group, geographic region), and time period. These reports must use CSSAP-validated computation methodology and must be accompanied by methodology documentation.
- **Annual Methodology Audits:** Independent third-party auditors approved by CSSAP must conduct annual audits of each covered platform's amplitude computation methodology, data sources, and reporting processes. Full audit reports must be submitted to ACIA; public summaries must be published with commercially sensitive details appropriately redacted.
- **Aggregate PSI Disclosure:** Platforms must publicly disclose aggregate PSI estimates for their user populations on a quarterly basis. Disclosure must be at the population level only — no individual-level cognitive state data may be disclosed — and must be accompanied by contextual explanation in plain language accessible to

non-specialist audiences.

- **Incident Reports:** Platforms must submit mandatory incident reports to ACIA when AA exceeds Red Zone thresholds ($AA > 0.60$) for more than 72 continuous hours across any population segment of 100,000 or more users. Incident reports must describe the amplitude breach, its likely causes, the corrective actions taken, and the timeline for return to compliant amplitude levels.
- **User-Level Amplitude Disclosure:** Platforms must, upon request, provide individual users with a summary of their personal estimated amplitude exposure history over the preceding 12 months, expressed in terms interpretable by a non-specialist, and with accompanying guidance on amplitude management options available to the user.

10. Implementation Guidelines

10.1 Platform Responsibilities

All platforms meeting the coverage threshold (≥ 1 million monthly active users in any CSF-adopting jurisdiction) bear the following mandatory responsibilities under the CSF regulatory framework. These responsibilities are designed to be implementable within a 24-month transition period for platforms that begin compliance preparation upon framework adoption, with phased implementation milestones at 6, 12, and 24 months.

Each covered platform must designate a **Cognitive Impact Officer (CIO)** — a senior executive with formal responsibility for amplitude compliance, reporting to the platform's board of directors and with direct authority over the platform's content delivery architecture. The CIO is the named regulatory contact for ACIA communications and bears personal professional accountability for the accuracy and completeness of mandatory regulatory reports. The CIO role is distinct from existing content policy or trust-and-safety functions, reflecting the specialised technical and regulatory knowledge required for amplitude governance.

Platforms must implement **real-time AA monitoring infrastructure** capable of computing $AA(t)$ for defined user population segments at time intervals no greater than one hour, and capable of generating alerts when AA values approach or breach regulatory thresholds. This infrastructure must be sufficiently robust to provide reliable monitoring

under peak load conditions and must maintain 99.5% uptime. AA monitoring systems must be distinct from platforms' content recommendation systems to ensure that monitoring cannot be compromised by optimisation processes that have incentive to minimise measured amplitude.

Platforms must establish **automated amplitude dampening mechanisms** — algorithmic interventions that are triggered automatically when AA approaches the Amber Zone upper boundary ($AA = 0.60$) — capable of reducing amplitude within defined time windows through the strategies specified in Section 10.3. Dampening mechanisms must be pre-approved by ACIA and must be demonstrated to operate as designed under audit conditions. Platforms must maintain **3-year longitudinal records** of all AA metrics, PSI estimates, reporting submissions, and dampening mechanism activations, available for inspection by ACIA auditors on request.

10.2 Measurement Protocols

Consistent and comparable AA measurement across platforms requires standardised computation methodology for each component of the AA equation. The following methodology standards are proposed by the CSF and subject to revision by CSSAP as measurement science matures.

M(t) — Content Magnitude Computation: $M(t)$ is computed from content emotional intensity scoring using validated psycholinguistic models applied to the text, image, video, and audio content of each delivered item. The scoring model must be trained on diverse, multi-language, multi-cultural content corpora and validated against independent affective intensity ratings. $M(t)$ values must be calibrated against the cross-platform normalisation standard published annually by CSSAP. Platforms must document their $M(t)$ computation model in full technical detail for CSSAP review.

V(t) — Velocity Computation: $V(t)$ is computed from server-side delivery logs, measured as the number of distinct content items delivered per minute per user, normalised against the CSSAP-defined reference delivery rate (currently proposed at 4 items per minute as the maximum consistent with epistemic processing at typical adult rates). $V(t)$ must be computed as a rolling 10-minute average to prevent short-term spikes from producing misleadingly high point measurements, but must also trigger real-time alerts when instantaneous velocity exceeds three times the rolling average.

F(t) — Frequency Resonance Factor Computation: $F(t)$ is computed from engagement elasticity models that quantify the correlation between content type and user session extension — specifically, the degree to which delivery of each content item increases the probability and duration of continued engagement. $F(t)$ values approaching 1.0 indicate

that the platform is delivering maximally engaging content for the individual user's current state, signalling high resonance targeting. Platforms must document their $F(t)$ computation methodology including the engagement elasticity model architecture and training data provenance.

P(t) — Targeting Precision Computation: $P(t)$ is computed from the platform's targeting algorithm depth score — the number of distinct personalisation dimensions (demographic, behavioural, inferred psychological, social network, temporal, geographic, device-specific, etc.) that are active in the content selection decision for a given user at a given moment, normalised against the maximum number of personalisation dimensions technically possible. Platforms must maintain a complete registry of all personalisation dimensions used in content selection and must report any additions to this registry to ACIA within 30 days of implementation.

10.3 Amplitude Reduction Strategies

When AA monitoring indicates that platform amplitude is approaching or exceeding regulatory thresholds, platforms must activate amplitude reduction strategies from the following evidence-informed toolkit. Strategies may be deployed individually or in combination, and must be selected on the basis of the specific AA component or components driving the threshold breach.

1. **Velocity Throttling** — targeting $V(t)$: Imposing minimum inter-item delivery intervals to reduce the rate at which new content items are presented to users. Velocity throttling is the most immediately effective amplitude reduction strategy because it operates directly on $V(t)$, which feeds multiplicatively into AA. Minimum inter-item intervals of 15–30 seconds, applied dynamically when AA breaches the Amber Zone, can produce substantial AA reductions without eliminating content access. Throttling protocols must be graduated — increasing interval length proportionally to the severity of the threshold breach — to avoid abrupt disruption of user experience that could itself constitute a destabilizing perturbation.
2. **Diversity Injection** — targeting $F(t)$: Introducing cognitively heterogeneous content into the delivery stream to reduce the resonance factor $F(t)$. Diversity injection operates by deliberately inserting content that is not highly resonant with the user's current attractor proximity — that is, content drawn from outside the user's characteristic engagement patterns. Effective diversity injection reduces $F(t)$ by preventing the persistent delivery of maximally resonant content that drives resonance cascade dynamics. Diversity injection protocols must be calibrated to avoid the opposite extreme of delivering entirely non-resonant content, which would reduce engagement to zero and constitute an excessively disruptive intervention.

3. **Precision Caps** — targeting $P(t)$: Limiting the number of active personalisation dimensions used in content selection to reduce the targeting precision coefficient $P(t)$. Precision caps are particularly effective at breaking the resonance cascade dynamic, as cascade induction depends on the platform's ability to deliver maximally resonant content with high precision to users whose CSVs are near basin boundaries. Precision caps may be applied platform-wide during periods of elevated aggregate AA, or selectively to user segments where individual $P(t)$ contributions to AA are disproportionately high.
4. **Intensity Normalization** — targeting $M(t)$: Algorithmic down-weighting of content items with extreme emotional intensity scores in the delivery ranking, reducing the average $M(t)$ of delivered content. Intensity normalization does not remove high-intensity content from the platform — it modifies its delivery priority to ensure that the average delivered intensity remains within compliant levels. This strategy is most appropriate when AA threshold breaches are primarily driven by $M(t)$ — that is, when the platform's content ecosystem contains a disproportionate volume of high-intensity content that is being delivered at high rates. Intensity normalization parameters must be disclosed to CSSAP to ensure they do not constitute covert content suppression on ideological grounds.
5. **Epistemic Rest Periods** — targeting cumulative exposure effects: Mandatory low-amplitude intervals inserted into user sessions at regular intervals, providing cognitive recovery time that limits cumulative basin erosion effects. The proposed standard is a minimum 10-minute low-intensity window for every 60 minutes of continuous high-amplitude session engagement. During epistemic rest periods, delivered content must have $M(t)$ below 0.20 and $V(t)$ below 0.30. Platforms may use these periods to present content categories with low perturbation potential (informational, educational, or creative content with low affective intensity) rather than simply interrupting the session. Epistemic rest periods are most effective at limiting cumulative exposure effects (the integral term in the basin erosion equation) when applied consistently throughout extended sessions.

Implementation Note: Strategy Sequencing

Amplitude reduction strategies should be deployed in a graduated sequence proportionate to the severity of the threshold breach. For minor Amber Zone exceedances, Diversity Injection and Precision Caps (strategies 2 and 3) are the preferred first-line interventions, as they address the resonance dynamics driving elevated $F(t)$ and $P(t)$ without disrupting the core content delivery experience. For sustained Amber Zone or initial Red Zone breaches, Velocity Throttling (strategy 1)

should be added. For sustained Red Zone breaches, all five strategies should be deployed simultaneously, and the mandatory incident report to ACIA must be filed within 72 hours.

11. Conclusion

11.1 Summary of Regulatory Need

This framework has made a systematic and mathematically grounded case that algorithmic content delivery systems operating at high amplitude constitute a measurable and growing threat to cognitive stability at population scale. The perturbation mechanisms described in Section 3 — amplitude escalation, resonance targeting, basin erosion, and cascade induction — are not speculative risks but the foreseeable outputs of engagement-optimised content delivery architectures that systematically favor high-amplitude content because high amplitude drives high engagement, and high engagement drives platform revenue. The mathematical models in Section 4 formalize these dynamics with sufficient precision to enable regulatory specification: the Algorithmic Amplitude equation provides a single comparable metric; the Cognitive State Vector provides a framework for individual-level monitoring; the basin erosion model demonstrates the cumulative and accelerating character of cognitive risk under sustained high-amplitude conditions; and the Population Stability Index provides the population-level aggregate measure against which regulatory compliance can be assessed. These models together demonstrate that unconstrained amplitude optimization is not merely producing a diffuse background of social disruption but is systematically shifting population-level cognitive state distributions toward negative attractor basins in ways that are mathematically predictable, empirically detectable, and — with sufficient early intervention — regulatorily manageable.

11.2 Call for Adoption

Three constituencies bear immediate responsibility for advancing the adoption of the Cognitive Stability Framework. **Regulators and legislative bodies** must prioritise the establishment of ACIA as an independent statutory authority with sufficient powers, resources, and technical expertise to implement effective amplitude oversight. The regulatory architecture described in Section 9 is designed to be adoptable within existing administrative law frameworks in most jurisdictions, and the CSF's mathematical specificity

enables the kind of performance-standard regulation that modern regulatory agencies are well equipped to administer. Establishing ACIA and mandating quarterly reporting as an initial step — before the full amplitude threshold enforcement regime is in place — would begin the essential work of building the empirical evidence base required for threshold calibration. **Platform operators** must recognise that voluntary early adoption of amplitude monitoring and reduction strategies is not merely an ethical obligation but a strategic imperative. The regulatory trajectory of algorithmic amplitude governance is clear; the question is only whether platforms engage constructively with the CSF framework during its formative period, when they can contribute meaningfully to threshold calibration and measurement protocol design, or whether they resist until mandatory requirements impose less favorable standards with less generous transition timelines. **The scientific community** must invest urgently in the empirical calibration of the mathematical models presented here — particularly the erosion rate constant λ , the attractor centroid coordinates (CSV_P* and CSV_N*), and the basin radii estimates that underpin the threshold specifications. The CSF provides a precise mathematical framework that can be operationalised in empirical research programs using existing methodologies from computational social science, cognitive psychology, and network science.

11.3 Final Remarks

The Cognitive Stability Framework presented in this document is explicitly a living document — a first-generation theoretical and regulatory architecture that will require iterative refinement as empirical evidence accumulates, measurement science matures, and platform architectures evolve. The mathematical models are calibrated on current best estimates of cognitive dynamical parameters; these estimates will improve substantially as dedicated empirical research programs are established. The threshold values and governance structures proposed here are grounded in the best available theoretical and analogical reasoning, but they carry significant uncertainty that only systematic empirical investigation can resolve. This uncertainty is not an argument for delay — the cost of waiting for perfect empirical calibration before acting is measured in the ongoing and accelerating erosion of the cognitive commons that amplitude-unregulated systems are producing in real time. It is, rather, an argument for establishing the governance architecture and reporting infrastructure now, so that the empirical calibration work can be conducted within a regulatory framework capable of translating scientific findings into protective action without legislative delay.

The cost of inaction is not abstract. It is measured, in the first instance, in the individual experiences of millions of people whose cognitive equilibrium is being systematically perturbed by systems they did not choose and cannot easily escape. It is measured, in the second instance, in the progressive erosion of the shared epistemic foundations — the

common factual ground, the intersubjective trust, the cooperative norms — upon which the institutions of democratic self-governance depend. And it is measured, ultimately, in the narrowing window of opportunity for effective intervention as basin erosion compounds and the cognitive infrastructure required for collective corrective action is itself progressively degraded by the very systems that intervention would constrain. The Cognitive Stability Framework is offered as the conceptual vocabulary, the mathematical scaffolding, and the institutional architecture for beginning that work before the window closes.

Appendix A: Notation Summary Table

The following table provides a complete reference for all mathematical symbols, functions, and indices used throughout this framework document. Symbols are listed alphabetically within each category.

Symbol	Definition	Units / Range
Core Variables		
AA(t)	Algorithmic Amplitude at time t — the product of M , V , F , and P ; the primary regulatory quantity of the CSF	Dimensionless, $[0, 1]$
AT	Amplitude Threshold — critical amplitude value above which basin displacement probability exceeds ϵ	Dimensionless, $[0, 1]$
AT_general	Amplitude Threshold for general adult population	0.60
AT_minor	Amplitude Threshold for users under 18 years	0.15
AT_vulnerable	Amplitude Threshold for identified vulnerable population groups	0.30
B(t)	Belief Coherence — first component of CSV; internal consistency of individual's belief structure at time t	$[0, 1]$
CSV(t)	Cognitive State Vector at time t — four-dimensional representation of individual cognitive configuration	Vector in $\mathbb{R}^4$;
CSV_N*	Negative Attractor Centroid — the cognitive state at	$[0.25, -0.70, 0.15,$

	the centre of the negative attractor basin	0.20]
CSV_P*	Primary Attractor Centroid — the cognitive state at the centre of the primary (generative) attractor basin	[0.80, 0.25, 0.70, 0.75]
E(t)	Emotional Valence — second component of CSV; net affective charge at time t	[−1, +1]
F(t)	Frequency Resonance Factor — degree of alignment between delivered content and user's attractor proximity; third component of AA	[0, 1]
M(t)	Magnitude — emotional/cognitive intensity of delivered content at time t; first component of AA	[0, 1]
N	Total population size — number of individuals in the regulated platform population	Positive integer
O(t)	Epistemic Openness — third component of CSV; capacity for belief revision at time t	[0, 1]
P(t)	Targeting Precision Coefficient — personalisation depth at time t; fourth component of AA	[0, 1]
PSI(t)	Population Stability Index at time t — proportion of population within the primary attractor basin Ω_P	[0, 1]; target ≥ 0.85
T(t)	Social Trust — fourth component of CSV; confidence in shared social reality at time t	[0, 1]
V(t)	Velocity — rate of content delivery at time t; second component of AA	Normalised, [0, 1]

Basin Parameters

r_N	Negative Attractor Basin radius — Euclidean distance defining the extent of the negative attractor basin Ω_N	Non-negative real
r_P	Primary Attractor Basin radius — Euclidean distance defining the extent of the primary attractor basin Ω_P ; the epistemic damping constant	Non-negative real
r_P(0)	Initial Primary Attractor Basin radius before algorithmic exposure; healthy epistemic damping capacity	Non-negative real

Ω_N	Negative Attractor Basin — set of CSV values within r_N of CSV_N^*	Subset of $\mathbb{R}^4$;
Ω_P	Primary Attractor Basin — set of CSV values within r_P of CSV_P^*	Subset of $\mathbb{R}^4$;

Model Parameters and Operators

ε	Socially acceptable displacement risk — probability tolerance for basin displacement per exposure event	0.05 (proposed)
λ	Basin erosion rate constant — empirically determined rate of epistemic damping decay under excess amplitude exposure	Non-negative real, population-specific
$[x]_+$	Positive-part operator — equals $\max(x, 0)$; ensures only excess amplitude above AT contributes to erosion	Non-negative real
$\mathbb{1}[\cdot]$	Indicator function — equals 1 if the condition in brackets is satisfied, 0 otherwise; used in PSI computation	$\{0, 1\}$
$\ \cdot\ $	Euclidean norm in $\mathbb{R}^4$; — distance measure used to define attractor basin membership in CSV space	Non-negative real
τ	Integration variable — dummy time variable in the basin erosion integral; ranges over $[0, t]$	Time dimension
t	Time — continuous independent variable; present moment of measurement or evaluation	Time dimension
T	Regulatory window — defined time horizon over which stability conditions must hold (30-day rolling window)	Time dimension; 30 days

Regulatory Thresholds and Indices

PSI_min	Minimum Population Stability Index — regulatory floor below which platforms are in breach of the fundamental stability requirement	0.85
AA_green	Green Zone amplitude ceiling — maximum AA consistent with minimal regulatory oversight	0.25
AA_amber	Amber Zone amplitude ceiling — maximum AA	0.60

	consistent with monitoring-only regulatory status	
AA_red	Red Zone threshold — AA value above which mandatory reduction protocols are triggered and enforcement action is initiated	> 0.60

Regulating Algorithmic Amplitude: A Cognitive Stability Framework

A Theoretical and Regulatory Model for Managing AI-Driven Cognitive Perturbation at Scale

Author: Matthew B Wareham | Date: July 2026 | Version 1.0

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threshold values are theoretical proposals subject to empirical calibration. The framework is intended for adoption,

adaptation, and further development by regulatory bodies, research institutions, and platform operators.